Inner Product Spaces

Math Foundations — Node 08

1 Motivation

In $\mathbb{R}^n$ the dot product $u \cdot v = \sum_i u_i v_i$ encodes both length, via $\|u\|^2 = u \cdot u$, and angle, via the relation $\cos\theta = (u \cdot v)/(\|u\|\,\|v\|)$. An *inner product space* is the abstraction that lets us carry these geometric notions into any vector space, including spaces of functions, sequences, and random variables. Linear algebra without an inner product gives us linear maps; with one, it gives us geometry.

2 Definitions

Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 1 (Inner product)**.** An *inner product* on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ satisfying, for all $u, v, w \in V$ and $\alpha \in \mathbb{F}$:

1. **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$.
2. **Linearity in the first argument:** $\langle \alpha u + w, v \rangle = \alpha \langle u, v \rangle + \langle w, v \rangle$.
3. **Positive definiteness:** $\langle u, u \rangle \geq 0$, with equality iff $u = 0$.

A vector space equipped with an inner product is called an *inner product space*.

Note that over $\mathbb{R}$, conjugate symmetry collapses to symmetry, and linearity in the first argument together with symmetry gives linearity in the second argument as well. Over $\mathbb{C}$, the second argument is *conjugate*-linear; this is sometimes called the "physicist's convention" when the conjugate sits in the first slot — we follow the mathematician's convention here.

Definition 2 (Induced norm)**.** The *induced norm* on an inner product space $(V, \langle \cdot, \cdot \rangle)$ is

$$\|v\| := \sqrt{\langle v, v \rangle}, \qquad v \in V.$$

Positive definiteness makes this well-defined and non-negative.

Definition 3 (Orthonormal set)**.** A finite or countable family $\{e_i\}_{i \in I} \subset V$ is called *orthonormal* if

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

3 The Cauchy-Schwarz inequality

Theorem 4 (Cauchy-Schwarz)**.** *Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F}$. For all $u, v \in V$,*

$$\big|\langle u, v\rangle\big| \;\le\; \|u\|\,\|v\|\,, \tag{1}$$

with equality if and only if u and v are linearly dependent.

Proof. If $v = 0$ both sides of (1) are zero and there is nothing to prove, so assume $v \neq 0$. The trick is to consider, for a scalar $t \in \mathbb{F}$ to be chosen, the non-negative quantity

$$0 \le \|u - tv\|^2 = \langle u - tv, u - tv\rangle.$$

Expanding using linearity in the first slot and conjugate linearity in the second,

$$\begin{aligned}\langle u - tv, u - tv\rangle &= \langle u, u\rangle - t\,\langle v, u\rangle - \bar{t}\,\langle u, v\rangle + |t|^2\langle v, v\rangle \\ &= \|u\|^2 - t\,\overline{\langle u, v\rangle} - \bar{t}\,\langle u, v\rangle + |t|^2\,\|v\|^2\,.\end{aligned}$$

Choose $t = \langle u, v\rangle / \|v\|^2$. Then $\bar{t}\langle u, v\rangle = |\langle u, v\rangle|^2 / \|v\|^2$ and $t\,\overline{\langle u, v\rangle} = |\langle u, v\rangle|^2 / \|v\|^2$, while $|t|^2\,\|v\|^2 = |\langle u, v\rangle|^2 / \|v\|^2$. Substituting,

$$0 \le \|u\|^2 - \frac{|\langle u, v\rangle|^2}{\|v\|^2}.$$

Multiplying through by $\|v\|^2 > 0$ and rearranging,

$$|\langle u, v\rangle|^2 \le \|u\|^2\,\|v\|^2\,,$$

and taking square roots gives (1).

For the equality case: equality forces $\|u - tv\|^2 = 0$, so by positive definiteness $u = tv$, i.e. u and v are linearly dependent. Conversely if $u = \lambda v$ then both sides of (1) equal $|\lambda|\,\|v\|^2$. ☐

Proposition 5 (Triangle inequality)**.** *For all $u, v \in V$, $\|u + v\| \le \|u\| + \|v\|$.*

Proof. Expand $\|u + v\|^2 = \|u\|^2 + 2\,\Re\,\langle u, v\rangle + \|v\|^2$, bound $\Re\,\langle u, v\rangle \le |\langle u, v\rangle| \le \|u\|\,\|v\|$ by Theorem 4, and take square roots. ☐

4 Examples

Example 6 (Euclidean space)**.** $V = \mathbb{R}^n$, $\langle u, v\rangle = \sum_{i=1}^n u_i v_i$. The induced norm is the usual Euclidean length, and Cauchy-Schwarz is the classical $\big(\sum u_i v_i\big)^2 \le \big(\sum u_i^2\big)\big(\sum v_i^2\big)$.

Example 7 (L^2 on the unit interval)**.** $V = C[0, 1]$ with $\langle f, g\rangle = \int_0^1 f(x)\,g(x)\,dx$. Cauchy-Schwarz becomes

$$\left|\int_0^1 f g\,dx\right| \le \left(\int_0^1 f^2\,dx\right)^{1/2} \left(\int_0^1 g^2\,dx\right)^{1/2},$$

a special case of Hölder's inequality with $p = q = 2$.

5 Where this leads

Cauchy-Schwarz is the inequality that lets us define $\cos\theta = \langle u, v\rangle / (\|u\|\,\|v\|) \in [-1, 1]$. The construction of orthonormal bases via Gram-Schmidt, and orthogonal projection onto a closed subspace, both rest on this single inequality. In ML, every similarity score that takes the form of a dot product — attention logits, contrastive loss similarities, cosine similarity — is a direct corollary.